

StrengthHub Online

The AI Coach — Capability Plan

What the coach can action and answer, feature by feature across the app

Planning document · illustrative, not exhaustive

Prepared 31 July 2026 · Draft for discussion

Purpose & how to read this

This is a plan, not a spec, and deliberately open-ended. The examples below map the obvious, high-value things the coach should handle across the app. They are **illustrative** — **certainly not the limit** of what the coach can do. Treat every list here as "including, but not limited to." The coach should be free to help with anything a knowledgeable, safety-aware personal trainer would, provided it stays inside the safety and data boundaries in the final section.

The coach is not limited to this list

New user questions, edge cases, and whole feature areas we haven't thought of yet are **in scope by default**. The right mental model is a capable human coach with full read-access to the user's data — these pages just document the paths we can name today.

As the app grows (new screens, new data, new integrations), the coach's reach should grow with it without a rewrite of this plan.

Three principles that shape every capability

- **Coach proposes, the safety engine performs & validates.** The coach never writes to a program directly. Every workout change is resolved by the deterministic generator, which re-clamps against the Safety Rules, and only then applies.
- **Two verbs per feature — "answer / advise" vs "action."** Advice is free-flowing; actions are concrete state changes and some require an explicit confirm step.
- **Everything is grounded in the user's real data.** The coach is handed a compact context block on every request: goal, body stats, experience, equipment, injuries, diet, targets, active busy period, streak, and workouts this week — and can read the rest of the app state as needed.

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1. Workouts & exercises

The richest area, and where the coach does real actioning. It reads the exercise database (which carries a plain-language "why it's in your program", "what it does", steps, common mistake and safety note per exercise), the user's split, and their progression state.

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Why today's session is what it is ("Why did I train chest today?") — explain the split, day type, and muscle rotation • What an exercise does, which muscles it targets, and why it's in the program • Form cues — the top 2–3 things that matter for a named lift; point to the in-workout clip • Why this sets / reps / RIR for this goal (reads the prescription grid) • Rest-time guidance, and why compounds rest longer than isolation • Frequency & rest-day questions, tuned to their days/week • Warm-up and how to choose a starting working weight • Tempo, unilateral ("per side"), and the time-based stations in 12-min quick workouts	Substitute an exercise for any coded reason: dislike, pain, no equipment, too hard, too easy, or "I want this specific lift" — returns the first eligible, safety-clamped option, never invents one Explain candidate swaps first ("2 alternatives that hit the same muscles with your kit"), then action the chosen one Pain-aware swaps exclude any exercise loading the flagged region — and escalate a genuine stop-symptom instead of swapping Reschedule a training day or edit the whole available-days set, then re-run the scheduler Handle a missed session — exempt, shift the week, fold key lifts into the next one, or re-plan Trigger a deload when a lift has stalled for weeks (drop ~10% and build back) Start the right session for the moment — full session or a 15-min quick session

2. Goals & program direction

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• "Am I on track for my goal?" — reads weight trend, streak, workouts/week and progression for a grounded yes/no • Trade-offs of changing goal (e.g. build-muscle → lose-fat): what stays, what shifts • Realistic timeline / rate-of-change for their goal and body stats • How their targets (calories, protein, days/week) were set and whether they still fit	Change training goal — re-selects split, re-budgets volume, re-prescribes, with a built-in easy transition week so intensity never spikes Adjust days/week or session length and regenerate around it Update goal weight / body targets and reflect it in projections Flag re-screening when a goal change needs health screening re-run before applying

3. Nutrition & food

Boundary: advise, don't author a plan

The coach **reviews, answers, and suggests** — it does not generate a structured meal plan or set macro targets from scratch. This boundary should be encoded as an explicit refusal path. Everything else about food is fair game.

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Free-text food-log review with a 0–10 quality score, goal-aware highlights and up to 3 improvements ("here's what I ate today") • Ask-anything food Q&A: protein, snacks, eating on a budget, fat loss, eating out, hydration, alcohol • Protein-target coaching — how much, spread across meals, cheap student sources • Budget-eating guidance and pointing to specific Budget Eats recipes that fit their macros/diet • Pre-/post-workout fuelling, and eating around exams • Dietary-restriction-aware suggestions (vegetarian / halal / dairy-free) • Evidence-based supplement questions (protein powder, creatine) • Turn the quick day-tags (takeout day, skipped meals...) into one useful nudge	Suggest & open a recipe from Budget Eats that fits — advising, not composing a plan Nudge to log water / nutrition when the day's numbers are low <i>(No plan generation, no from-scratch macro-setting — see boundary above)</i>

4. Progress, stats & PRs

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Interpret the weight trend ("scale bounced — look at the 4-week trend") • Explain and celebrate a real, freshly-logged PR • Read the estimated-1RM series for a lift and say whether it's climbing, flat, or stalling • Explain weekly volume trend and training consistency • "Why isn't my bench moving?" — cross-reference progression, sleep and adherence before advising	Offer to share a PR to the cohort (with confirm) Point to the exact Progress card / metric behind an answer

5. Recovery, wellbeing & habits

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Soreness vs joint pain triage (DOMS normal; sharp joint pain → train around it / see a pro) • Sleep coaching tied to their sleep target and this week's average	Nudge to log a habit (water / sleep / steps) when below target Suggest & start a shorter session on a low-energy day

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Fatigue / low-motivation days — "shrink the task", rest-is-training, streak protection • Hydration, steps and mindfulness targets • Stress / exam-season load management	

6. Plan around your life — busy periods & exams

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Explain how training bends during a declared busy period (pause / maintenance / keep-moving / fewer days / deload) • Reassure during exam mode ("three key lifts + sleep is a full win")	Turn on Exam mode and explain the automatic lightening Declare a planned absence (travel, moving) so those days never count as missed Choose the right period mode for their situation and apply it

7. New to the gym & app help

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Answer beginner-track questions and recommend the next lesson (gym / app / mindset / machine) • "How do I use X in the app?" — where to log, swap, or find form clips • Gym etiquette, picking a starting weight, what RIR means	Open the relevant lesson or app surface (beginner lesson, exercise detail, etc.)

8. Community

Answer / advise	Action (coach proposes → safety engine performs & re-clamps)
• Explain challenges, leaderboards and belonging-scoped competition (campus / dorm / society) • Suggest a training-partner match and explain the match %	Draft a PR / progress post for the user to review and publish Suggest joining a challenge or a partner connection <i>All outward posting is gated behind an explicit confirm</i>

9. Cross-cutting — what the coach reads, and hard boundaries

Context the coach always has

Handed to the coach on every request: name / age / sex, goal, height / weight / target, experience, days & minutes per session, equipment, injuries, diet + budget flag, calorie & protein targets, any active busy period, motivation, current streak, and workouts this week.

Context gaps worth adding

Recent PRs · current program day · last session's completion · sleep & water 7-day averages · per-lift plateau flags.

Non-negotiable boundaries to encode

Never writes to a program directly — always proposes to the deterministic engine, which re-clamps against the Safety Rules.

Never invents an exercise, weight or macro outside the seeded database / grid.

Safety routing runs first — crisis / disordered-eating / off-topic / under-18 checks precede any coaching or the daily limit. The coach ships gated OFF until independent review.

No medical diagnosis or personalised medical / physio advice — triage and refer.

No meal-plan generation or from-scratch macro-setting (Section 3 boundary).

Outward / irreversible actions (posting, connecting a partner) require an explicit user confirm.

10. Open questions before this becomes a build spec

- **Action scope for v1:** should the coach actually action swaps / goal-changes / reschedules, or only recommend and deep-link the user to do it themselves at first? (The engine supports full actioning today; the question is trust and UX.)
- **Confirm steps:** which actions get a one-tap confirm vs happen silently — and how the user undoes an actioned change.
- **Next artefact:** turn this into a capability matrix (feature × answer/action × backend module × confirm-required × safety-gate) so it maps straight to tickets.

Reminder

This document is a starting map, not a fence. The coach's job is to be genuinely helpful across the whole app — these are the paths we can name today, and the list is meant to grow.